UNIVERSITY OF WATERLOO

Beyond Labels: Unsupervised Anomaly Detection for Financial Fraud

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The Problem

- Fraudulent transactions are rare, evolving, and often poorly represented by reliable labels.
- A model that only learns from past confirmed fraud misses new fraud patterns and mislabeled cases.
- This motivates an unsupervised approach: score every transaction by how unusual it looks, without using fraud labels during training.

[X.X]%

of transactions in the IEEE-CIS dataset are labeled fraudulent

Kaggle competition page

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2. Data
3. Methods
4. Results
5. Comparison
6. Conclusion

Research Questions

RQ1 **How effectively can Isolation Forest and LOF identify fraud?**

How effectively can Isolation Forest and LOF identify fraud in IEEE-CIS without using labels during training?

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RQ2 **How does detection quality compare?**

How does detection quality (precision-recall AUC, ROC AUC, precision at a fixed review budget) compare between the two methods?

Research Questions

RQ1 **How effectively can Isolation Forest and LOF identify fraud?**

How effectively can Isolation Forest and LOF identify fraud in IEEE-CIS without using labels during training?

RQ2 **How does detection quality compare?**

How does detection quality (precision-recall AUC, ROC AUC, precision at a fixed review budget) compare between the two methods?

RQ3 **How do cost and sensitivity compare?**

How do computational cost and sensitivity to preprocessing/contamination assumptions compare between the two methods?

Dataset: IEEE-CIS Fraud Detection

590,540

transactions

`train_transaction.csv`

401

numeric features

after cleaning, see `00_data_cleaning.ipynb`

~3.5%

fraud rate

used as the contamination anchor throughout

> **Preprocessing:** [describe median imputation, any encoding, what was dropped].
>
> Labels withheld during `fit`, used only to evaluate scores after the fact.

Method: Isolation Forest

1

Core idea

Randomly partitions the feature space with splits. Anomalies isolate in fewer splits than normal points.

2

Parameters tuned

`n_estimators`, `max_samples`, `contamination`

best values from Task 4

3

Result

[One-sentence takeaway from the tuned baseline]

[INSERT: random-partitioning / isolation-path diagram]

Method: Local Outlier Factor

1

Core idea

Compares a point's local density to its neighbors'. A point in a notably sparser neighborhood scores high.

2

Parameters tuned

`n_neighbors`, `contamination`, subsample size

best values from Task 6

3

Result

[One-sentence takeaway from the tuned baseline]

[INSERT: local-density / neighborhood diagram]

Isolation Forest Results

[INSERT: score distribution plot Task 3/4 notebook]

METRIC	BASE	TUNED
ROC-AUC	[-]	[-]
PR-AUC	[-]	[-]
Precision @ 3.5%	[-]	[-]
Recall @ 3.5%	[-]	[-]
Runtime	[-]	[-]

LOF Results

[INSERT: score distribution plot Task 5/6 notebook]

METRIC	BASE	TUNED
ROC-AUC	[-]	[-]
PR-AUC	[-]	[-]
Precision @ 3.5%	[-]	[-]
Recall @ 3.5%	[-]	[-]
Runtime	[-]	[-]

Side-by-Side Comparison

	ISOLATION FOREST (TUNED)	LOF (TUNED)
ROC-AUC	[–]	[–]
PR-AUC	[–]	[–]
Precision @ fixed budget	[–]	[–]
Recall @ fixed budget	[–]	[–]
Runtime (same sample size)	[–]	[–]

[One line: which method ranks fraud higher overall, and what trade-off drives the final pick]

Strengths & Weaknesses

ISOLATION FOREST

- + strength
- + strength
- – weakness

LOCAL OUTLIER FACTOR

- + strength
- + strength
- – weakness

Limitations and Next Steps

LIMITATIONS

- Label noise / unreliable ground truth.
- Sensitivity to contamination/threshold choices.
- LOF's cost in 401 dimensions.

NEXT STEPS

- Autoencoder stretch goal, if time permits.
- Any follow-up analysis flagged in Task 7.

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Takeaways

RQ1: detection without labels

[Headline answer]

Takeaways

RQ1: detection without labels

[Headline answer]

RQ2: quality comparison

[Headline answer]

Takeaways

RQ1: detection without labels

[Headline answer]

RQ2: quality comparison

[Headline answer]

RQ3: cost & sensitivity

[Headline answer]

Thank You

Questions?



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PROGRAM	Women in Mathematics Directed Research Program
UNIVERSITY	University of Waterloo

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[2] Breunig, M. M., Kriegel, H.-P., Ng, R. T., & Sander, J. (2000). LOF: Identifying Density-Based Local Outliers. *ACM SIGMOD*.

[3] IEEE-CIS Fraud Detection dataset. Kaggle competition page.

[4] Psychoula, I., Gutmann, A., Mainali, P., Lee, S. H., Dunphy, P., & Petitcolas, F. A. P. Explainable Machine Learning for Fraud Detection. OneSpan Innovation Centre.

[5] Waspada, I. et al. Performance Analysis of Isolation Forest Algorithm in Fraud Detection of Credit Card Transactions.

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Appendix Contents

01 Feature list and preprocessing

02 Metric definitions

03 Fine-tuning: values tried

04 LOF runtime benchmark

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Feature List and Preprocessing

CATEGORY	FEATURES
Transaction	amount, product code, card type, ...
Identity	device type, browser, ...
Address / distance	billing/shipping region, distance fields, ...
Counting features	C1–C14 style aggregate counts
Time-delta features	D1–D15 style, days since event
Match features	M1–M9 style, yes/no match flags

401 numeric features used after cleaning. [Fill in from 00_data_cleaning.ipynb / 01_eda.ipynb]

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Metric Definitions

ROC-AUC Probability a randomly chosen fraudulent transaction scores higher than a randomly chosen legitimate one. 0.5 = chance, 1 = perfect.

PR-AUC Area under the precision-recall curve. More informative than ROC-AUC under heavy class imbalance, relevant given the low fraud rate here.

Precision Share of the top-k flagged transactions that are actually fraud.

Recall Share of all fraud caught within the top-k flagged transactions.

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Fine-Tuning: Values Tried

SETTING	ISOLATION FOREST	LOF	IF CHOSEN	LOF CHOSEN
Ensemble / neighborhood size	n_estimators: 50, 100, 200, 500	n_neighbors: 5, 20, 50, 100	[–]	[–]
Sample size	max_samples: default, 128, 512, full	subsample: 10k, 30k, 50k rows	[–]	[–]
Contamination	0.01, 0.02, ~0.035, 0.05, 0.1	0.01, 0.02, ~0.035, 0.05, 0.1	[–]	[–]

Same true fraud rate (~3.5%) used as a contamination anchor for both methods, see Task 4 and Task 6.

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LOF Runtime Benchmark

20.24 s
30,000 rows

56.29 s
50,000 rows

~2 hours
590,540 rows
extrapolated

Brute-force distance computation scales roughly $O(n^2)$ with 401 features, this is why Task 5/6 used a subsample instead of the full dataset.